

Zain Ul Abidin

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Education

Bachelor of Engineering in Computer Systems Engineering

2022–2026

Sukkur IBA University, Sukkur, Sindh, Pakistan

CGPA: 3.80 / 4.00 | Rank: Top 3 in cohort

Bachelor's Thesis: "Machine Learning-Based Subsurface Defect Detection in CFRP Using Optical Pulsed Thermography" (Supervisor: Dr. Junaid Ahmed)

Developed an automated pipeline (Detection → Segmentation → Depth Estimation → 3D Reconstruction) for inspecting carbon fiber composites, pioneering a novel 3-channel image fusion method.

Research Interests

Computer Vision | Deep Learning | Non-Destructive Testing | Infrared Thermography | Structural Health Monitoring | Signal Processing | Image Processing | Intelligent Sensing Systems | Autonomous Multi-Agent Systems

Technical Skills

Programming & Databases: Python, C, C++, MATLAB, SQL, SQL Server, MySQL, SQLite

Machine Learning & Deep Learning: PyTorch, TensorFlow, Keras, scikit-learn, XGBoost, LLMs, SHAP

Computer Vision & Signal Processing: OpenCV, YOLO, Faster/Mask R-CNN, ViT, RT-DETR, Image Processing

Frameworks & Tools: ASP.NET, Streamlit, Flask, n8n, Edge AI (TF Lite, ESP32), Git, Linux

Publications

Published

- **Zain Ul Abidin**, Haseeb, H. Memon, M. Younis, J. Ahmed. "Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers." *International Journal of Multidisciplinary Conference Proceedings (IJMCP)*, Vol. 2(1), 2025. DOI: [10.61503/Ijmcpr.v2i1.181](https://doi.org/10.61503/Ijmcpr.v2i1.181)
- H. Memon, **Zain Ul Abidin**, J. Ahmed, A. Siddiqua. "Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning." *2026 5th International Conference on Computing, Mathematics and Engineering Technologies (iCoMET)*, Sukkur, Pakistan, IEEE, 2026, pp. 1–6. DOI: [10.1109/iCoMET69771.2026.11591843](https://doi.org/10.1109/iCoMET69771.2026.11591843)

Under Review

- **Zain Ul Abidin**, J. Ahmed, D. Wang, G. Tian. "Metal Defect Detection in Pulsed Thermographic Imaging via Temporal Eigen Projection and Robust Low-Rank Tensor Completion Model." *Nondestructive Testing and Evaluation*, Taylor and Francis, 2025.
- J. Ahmed, **Zain Ul Abidin**, G. Tian, D. Wang. "Eddy Current Pulsed Thermography based Efficient Defect Detection in Metals via Keyframe Extraction and Accelerated Sparse Decomposition." *NDT & E International*, Elsevier, 2025.
- P. Zhiwen, F. Sahito, A. Kumar, J. Ahmed, **Zain Ul Abidin**. "Improving Image Super-Resolution: A Hybrid Approach Using DT-CWT and Transformer Networks." *Image and Vision Computing*, Elsevier, 2025.
- **Zain Ul Abidin**, H. Memon, J. Ahmed. "Three-Channel Thermographic Fusion for Architecture-Independent Defect Detection and Segmentation in Carbon Fibre Reinforced Polymer Composites Using Deep Learning." *Engineering Applications of Artificial Intelligence*, Elsevier, 2026.
- H. Memon, **Zain Ul Abidin**, J. Ahmed. "Automated Depth Estimation and 3D Reconstruction of CFRP Subsurface Defects via Optical Pulsed Thermography." *Transactions on Instrumentation and Measurement*, IEEE, 2026.
- R. Kumar, W. Ali, J. Ahmed, N. I. Ali, S. Usman, **Zain Ul Abidin**. "AgentForge: Execution-Grounded Multi-Agent LLM Framework for Autonomous Software Engineering." *Automated Software Engineering*, Springer Nature, 2026.

Conference Presentations

- **International Conference on Information Science and Communication Technology (ICISCT 2025)** – "Mitigation of DDoS Attacks with Machine Learning, Deep Learning, and Transformers" (Paper Presentation). DHA Suffa University, Karachi, Pakistan. May 2025.
- **International Conference on Computing, Mathematics & Engineering Technologies (iCoMET 2026)** – "Debond Detection in CFRP Structures using Optical Pulse Thermography and Deep Learning" (Paper Presentation). Sukkur IBA University, Pakistan. February 2026.

Research & Development Projects

- **AgentForge: Autonomous Multi-Agent Research Pipeline** 2026
Engineered a 23-stage autonomous pipeline using specialized AI agents to independently process topics, generate code, execute experiments, and validate results.
- **KL-Divergence-Driven Online Sensor Relocation for Non-Stationary Barrier Coverage** 2026
Implemented sliding-window kernel density estimation and greedy sensor relocation, improving the mean detection rate from 65.16% to 87.30%.
- **X-ray Prohibited Item Detection in Airport Inspection** 2026
Developed an automated system using deep learning architectures to identify and segment prohibited items within complex, cluttered X-ray security scans.
- **Cross-Modal Fake News Detection (CMM-DSE)** 2026
Designed a multimodal AI system combining 2D FFT image features and text analysis via Dempster-Shafer reasoning to quantify uncertainty.
- **AI-Based Android Controlled Vision Robot** [Code] 2025
Programmed an ESP32-CAM with TensorFlow Lite for on-device real-time object detection to demonstrate edge AI deployment.
- **Real-Time Speech Emotion Recognition via LSTM** [Code] 2025
Trained an LSTM classifier on the TESS dataset with noise injection and time-shifting augmentation to achieve real-time inference on live audio.
- **Gold Price Forecasting Dashboard using Multimodal Ensemble** [Code] 2025
Created an interactive dashboard visualizing an ensemble of XGBoost, LSTM, and Random Forest models trained on historical market data.
- **Speech Enhancement via Spectral Subtraction and Wiener Filtering** [Code] 2025
Built a real-time pipeline that estimates background noise from degraded speech and applies Wiener filtering to recover intelligibility.
- **Hand Gesture Recognition and Voice Conversion** [Code] 2024
Designed a CNN and SVM pipeline for sign language gesture classification that converts gestures to synthesized speech in real-time.

Industrial Experience

- **Artificial Intelligence Intern** | *Mirai School of Technology, Remote* Jul 2025–Aug 2025
Designed and deployed a Conversational Order Assistant using agentic AI workflows to automate order capture, validation, logging, and email confirmation.
- **Software Engineering Intern** | *SEPCO, Sukkur, Sindh* Jun 2025–Jul 2025
Built the Mute Meter Monitoring Dashboard with GIS zone filters and bulk data processing capabilities, reducing manual field inspection workload by 40%.

Honours and Awards

- **Prime Minister's Laptop Award** (2024) – National merit-based award for academic excellence, Government of Pakistan.
- **Sindh Educational Endowment Fund (SEEF) Scholarship** (2022–2026) – Fully merit-based undergraduate scholarship, Government of Sindh.

Specializations & Certifications

- Machine Learning Specialization – [DeepLearning.AI / Coursera](#)
- Mathematics for Machine Learning and Data Science – [DeepLearning.AI / Coursera](#)
- Deep Learning Specialization: Advanced AI, Hands-on Lab – [Udemy](#)
- Large Language Models – [Udemy](#)
- Google Prompting Essentials Specialization – [Google / Coursera](#)
- Google IT Support – [Google / Coursera](#)

References

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